Kiran Teja Kandiboyina

Palakol, India | +91 8247432020 | kiranteja.kandiboyina@gmail.com
https://www.linkedin.com/in/kiran-teja-kandiboyina | https://github.com/kiranTejaK

Professional Summary

Backend Engineer with nearly 3 years of professional experience building production REST APIs and backend services using Python and FastAPI. Experienced in database design, SQL query optimization, Redis caching, asynchronous document-processing workflows, AWS integrations, JWT authentication, role-based access control, payment integrations, automated testing, and CI/CD. Delivered 100+ RESTful APIs and reusable backend components. Currently developing AI backend engineering capabilities through a project focused on LLM integration, Retrieval-Augmented Generation (RAG), embeddings, vector search, and hybrid information retrieval.

Technical Skills

Programming and Backend Development: Python, FastAPI, REST API Development, SQLAlchemy, Pydantic V2, Asynchronous Programming, Slim Framework
Databases and Data Access: MySQL, PostgreSQL, SQL, Stored Procedures, Transactions, Query Optimization, Indexing Strategies, Elasticsearch
Caching and Performance: Redis, Cache-Aside Pattern, Cache-Key Design, TTL Strategies, API Performance Optimization
AI and LLM Engineering – Project Development: Generative AI, Large Language Models (LLMs), LLM API Integration, Retrieval-Augmented Generation (RAG), Embeddings, Text Chunking, Vector Search, Semantic Search, Hybrid Search, pgvector, Full-Text Search, Reciprocal Rank Fusion (RRF)
Cloud and DevOps: AWS EC2, Amazon S3, AWS SQS, IAM, Docker, Docker Compose, GitHub Actions, Traefik, Linux, Git, CI/CD Pipelines
Testing and Observability: Pytest, Unit Testing, Integration Testing, Mocking, Structlog, Structured JSON Logging
Security and Integrations: JWT Authentication, Role-Based Access Control (RBAC), Stripe Webhook Verification, SMTP
Additional Technologies: React, JavaScript, HTML, CSS

Professional Experience

Backend Developer – KloudWorx Technologies Pvt Ltd **Apr 2023 – Jan 2026**
Enterprise HRTech, Digital Onboarding and E-Signature Platforms

- Developed and maintained production backend services primarily using Python and FastAPI for enterprise HRTech, digital onboarding, e-signature, and government platforms, with additional backend development using Slim Framework.
- Designed, developed, and delivered 100+ RESTful APIs, collaborating with frontend and database engineers to implement production features.
- Built asynchronous, event-driven document-processing workflows using AWS SQS and structured messages for downstream processing.
- Implemented secure Amazon S3 file-management workflows using pre-signed URLs for controlled document uploads and downloads.
- Optimized MySQL stored procedures, indexing strategies, and query execution plans to improve database performance under concurrent production workloads.
- Improved API response times by 25–30% using Redis cache-aside caching, optimized cache keys, and TTL-based cache invalidation.
- Evaluated search solutions and designed an on-premise Elasticsearch architecture for complex nested JSON search requirements.
- Implemented JWT authentication, Role-Based Access Control (RBAC), Stripe payment integrations, and secure, idempotent webhook processing.
- Introduced structured JSON logging with Structlog and automated unit and integration tests with Pytest to improve observability and deployment confidence.
- Developed a reusable FastAPI starter template with authentication, Redis caching, MySQL connectivity, stored-procedure execution, AWS S3, SMTP, and centralized exception handling.

Projects

SentinelRAG – Multi-Tenant AI Gateway and Hybrid Retrieval Platform **GitHub Repository**
In Progress

- Developing a Python and FastAPI-based AI gateway architecture to centralize LLM API integration, tenant-aware request processing, and provider routing.
- Designing a Retrieval-Augmented Generation (RAG) pipeline involving document ingestion, text chunking, embeddings, PostgreSQL/pgvector vector search, and PostgreSQL full-text retrieval.
- Designing hybrid retrieval using semantic and lexical search, with Reciprocal Rank Fusion (RRF) to combine ranked results.
- Planning Redis-based exact-match and semantic caching, tenant-aware rate limiting, provider failure handling, structured logging, and automated integration tests.
- **Technologies:** Python, FastAPI, PostgreSQL, pgvector, Redis, RAG, LLM APIs, Embeddings, Hybrid Search, Docker, Pytest

DOit – Workflow Management Platform **Live Demo**

- Designed and built a task and workspace management application using FastAPI, PostgreSQL, Redis, and React.
- Implemented JWT authentication and RBAC for secure multi-workspace authorization using reusable service layers and SQLAlchemy query patterns.
- Improved API performance through Redis cache-aside caching, optimized cache keys, and TTL-based invalidation.
- Containerized the application using Docker Compose and Traefik and automated deployment with GitHub Actions and AWS EC2.
- **Technologies:** FastAPI, Python, PostgreSQL, Redis, SQLAlchemy, React, Docker Compose, Traefik, GitHub Actions, AWS EC2

Education

M.Sc. Chemistry – Adikavi Nannaya University **2017 – 2019**

CCBP Intensive Program – NxtWave **2022 – 2023**
Hands-on Full-Stack Development Program

Teaching Experience

Teaching Assistant – CCBP Intensive, NxtWave **Aug 2022 – Apr 2023**
Supported technical doubt resolution and concept clarification

Certifications

Web Development with Python and Flask – Jovian